

ChipTherm: Physics-Guided Thermal Prediction Across Chiplet Package Families

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I. PROBLEM AND MOTIVATION

Thermal analysis is critical in chiplet-based and heterogeneous packages, where placement, power density, and package properties produce strongly coupled temperature fields. Reference simulators such as HotSpot [1] provide detailed maps, but their runtime limits repeated evaluation during placement and design-space exploration. Learned surrogates can accelerate this process. However, reported accuracy on unseen samples often conflates two different settings. A model may interpolate to new workloads within a known package family or generalize to an entirely unseen package family.

This distinction matters because a surrogate may perform well when the package structure is represented during training yet degrade substantially under package-level distribution shift. ChipTherm targets this challenge through separate seen- and unseen-family evaluation protocols and a fast, physics-guided source-decomposition predictor for repeated screening before detailed thermal verification.

II. BACKGROUND AND RELATED WORK

Compact simulators such as HotSpot support early-stage thermal estimation, while recent EDA frameworks combine analytical models and learned surrogates for thermal-aware optimization [1], [2]. Operator-learning methods further reduce simulation cost by learning mappings from power, geometry, and boundary conditions to spatial temperature fields. DeepOHeat applies physics-aware operator learning to 3D-IC simulation [3], while ThePlace embeds an FNO-based simulator in placement optimization [4]. Related architectures such as FNO, U-FNO, and SAU-FNO provide spectral modeling with multiscale or attention-based feature extraction [5]–[7].

Existing learned surrogates largely predict the temperature field directly from package-level inputs [3], [4], [7]. ChipTherm instead factorizes prediction into shared per-source unit-power thermal-response estimation, explicit power-scaled superposition, and decomposed package-level correction. This formulation imposes source-wise additive structure, reuses one source model across chiplets, and leaves the residual network to learn package-scale offsets and localized interactions.

III. APPROACH AND UNIQUENESS

As illustrated in Fig. 1, ChipTherm predicts package temperature through a two-stage, physics-guided decomposition. For each chiplet s , a shared source-response network processes

a 17-channel, full-package 64×64 representation encoding package occupancy and chiplet type, spatial coordinates, the selected-source mask and power density, source-relative geometry, and package-boundary distances. The network predicts an effective source-to-grid unit response $\hat{Z}_s(\mathbf{x})$ in K/W. Each response is multiplied by the corresponding chiplet power and summed with ambient temperature:

$$\hat{T}_{\text{base}}(\mathbf{x}) = T_{\text{amb}} + \sum_s P_s \hat{Z}_s(\mathbf{x}).$$

This learned source-superposition baseline explicitly preserves source-wise power scaling and additive thermal contributions rather than requiring one network to infer the complete package field directly.

A 2.19M-parameter residual CNN predicts package effects not captured by source superposition. Its spatial input concatenates a 33-channel package representation with the normalized source-superposition baseline, yielding 34 spatial channels. A separate 15-dimensional metadata vector summarizes package dimensions, power statistics, chiplet count, occupancy, and cooling geometry. Global/local multiscale features are conditioned by this metadata through FiLM before predicting an effective-resistance correction $\widehat{\Delta R}_{\text{eff}}$ and an explicitly zero-mean spatial residual $\hat{S}_0(\mathbf{x})$:

$$\hat{T}(\mathbf{x}) = \hat{T}_{\text{base}}(\mathbf{x}) + P_{\text{tot}} \widehat{\Delta R}_{\text{eff}} + \hat{S}_0(\mathbf{x}), \quad \langle \hat{S}_0 \rangle = 0.$$

This decomposition separates package-wide temperature shifts from localized interaction patterns.

Using HotSpot, we generate 10,000 temperature maps from 50 structurally distinct package families and 200 power workloads per family. Families vary in chiplet count, geometry, composition, occupancy, and placement; packages contain 6–64 chiplets and span 1088–4480 mm². All families intentionally share a common material stack and cooling boundary condition, isolating variation in package structure and workload. Under this controlled thermal context, 40 families are used for training, 5 unseen families for model selection, and 5 additional unseen families for final testing; a separate workload split within the training families measures seen-family interpolation.

IV. RESULTS AND CONTRIBUTIONS

Table I compares ChipTherm with neural-operator surrogates under identical evaluation protocols. ChipTherm

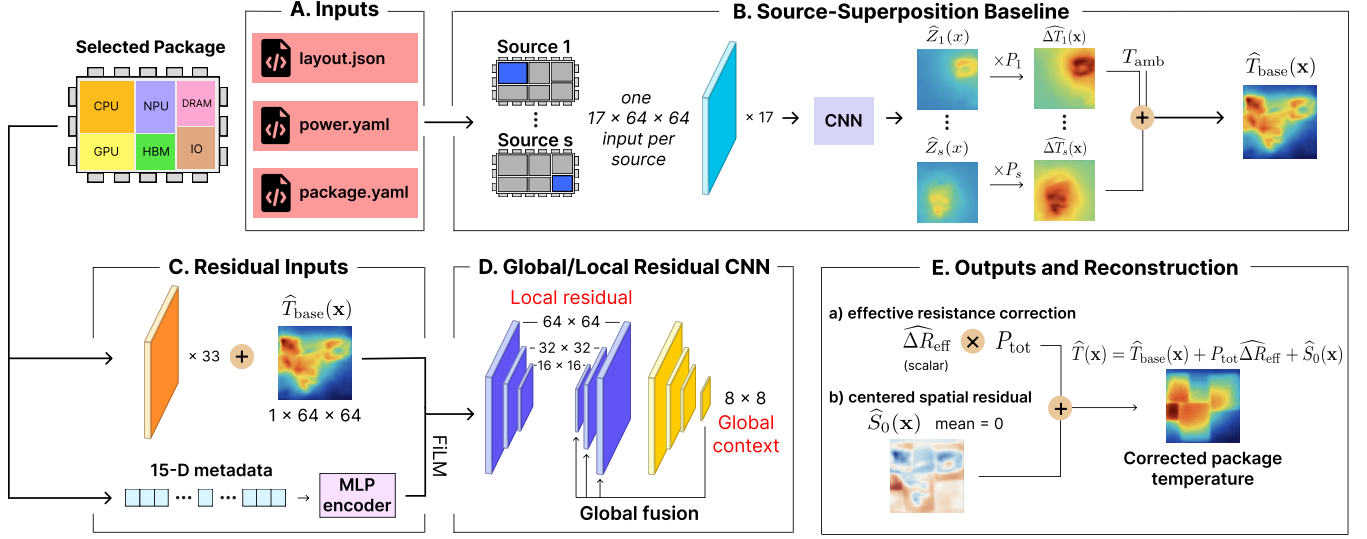


Fig. 1. ChipTherm inference pipeline. A shared source model predicts per-chiplet effective unit responses, which are power-scaled and summed to form a source-superposition baseline. A global/local residual CNN then predicts a package-wide effective-resistance correction and a centered spatial residual.

TABLE I
FULL-MAP ERRORS AGAINST HOTSPOT-GENERATED REFERENCE TEMPERATURES.

Model	Params (M)	Seen Family		Unseen Family	
		MAE (K)	RMSE (K)	MAE (K)	RMSE (K)
FNO	2.389	0.375	0.581	1.494	2.447
U-FNO	4.019	0.338	0.561	1.754	2.821
SAU-FNO	4.023	0.528	0.754	1.545	2.398
ChipTherm	2.664	0.150	0.284	1.331	2.486

TABLE II
CHIPTHERM PIPELINE-STAGE ABLATION USING FULL-MAP MAE.

Configuration	Seen MAE (K)	Unseen MAE (K)
Direct CNN	0.370	1.767
Source-superposition baseline	1.388	1.668
Full ChipTherm	0.150	1.331
Reduction vs. Direct CNN	59.5%	24.7%

achieves the lowest average full-map MAE in both evaluation regimes and the lowest seen-family RMSE. On seen package families, it reaches 0.150 K MAE and 0.284 K RMSE, reducing MAE by 55.6% relative to the strongest seen-family baseline, U-FNO. Under strict family holdout, ChipTherm reaches 1.331 K MAE, improving by 10.9% over FNO at 1.494 K. SAU-FNO obtains the lowest unseen-family RMSE, 2.398 K, compared with 2.486 K for ChipTherm, indicating that ChipTherm’s MAE advantage does not extend uniformly to larger spatial errors.

The gap between seen- and unseen-family errors shows that seen-family workload interpolation is substantially easier than generalization to package structures absent from training. ChipTherm narrows this gap in average full-map error, but unseen-family generalization remains challenging.

Table II compares the effects of source superposition and

learned package correction. The source-superposition baseline alone modestly improves unseen-family MAE over direct CNN prediction, from 1.767 to 1.668 K, but remains insufficient for accurate seen-family prediction. Adding the residual stage reduces MAE to 0.150 K and 1.331 K in the seen and unseen regimes, respectively. The two stages are therefore complementary: source-wise superposition supplies a structured initial estimate, while the effective-resistance and centered spatial corrections recover package-wide offsets and localized interaction patterns.

ChipTherm contains 2.664M parameters. Although slightly larger than FNO, it uses approximately 34% fewer parameters than U-FNO and SAU-FNO while achieving lower MAE in both regimes. Uncached inference takes 23.45 ms per package on an NVIDIA RTX A6000, a 210.8 $\times$ speedup over the recorded HotSpot runtime in our reference setup.

Overall, ChipTherm contributes a family-structured evaluation protocol and a compact source-superposition residual architecture that improves average full-field accuracy while exposing the remaining difficulty of generalization to unseen package families.

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